

MODULE 14

DTOs, Validation and API Contracts

Design effective Data Transfer Objects, implement robust validation, and define clear API contracts.







MODULE 14 OVERVIEW

Designing strong, predictable, and reliable APIs — DTOs, validation, and API contracts.

- Learn how to design effective Data Transfer Objects, implement robust validation, and define clear API contracts to build secure, maintainable, consumer-friendly services.

DURATION & LAB



1 Hour Theory

DTOs, Bean Validation, API contracts.



2 Hours Lab

DTOs and Validation.



Scenario

Build validated Customer DTOs for Northstar CRM.

DTO DESIGN

Purpose-driven, simple, immutable, version-friendly

VALIDATION

Jakarta Validation (formerly JSR 380) at the API boundary

CONTRACTS

Consistent, versioned agreements with consumers

KEY TAKEAWAY Well-designed DTOs, strong validation, and clear API contracts are the foundation of reliable, secure APIs.



WHY DTOS MATTER

DTOs act as a contract between your API and its consumers — build APIs that are secure, maintainable, and consumer-friendly.

WITHOUT DTOS

- Over-exposure of sensitive fields (passwords, audit data)
- Tight coupling — consumers depend on internal models
- Poor performance — unnecessary/heavy data
- Difficult to evolve without breaking clients

VS

WITH DTOS

- Data security — expose only what's needed
- Loose coupling — a stable contract
- Better performance — smaller payloads
- Easier evolution and versioning

COMMON USE CASES FOR DTOS



Mobile Apps

Lightweight data tailored for constrained clients.



Public APIs

Expose safe, versioned data to external partners.



Multiple Views

Different DTOs for list, detail, or report views.

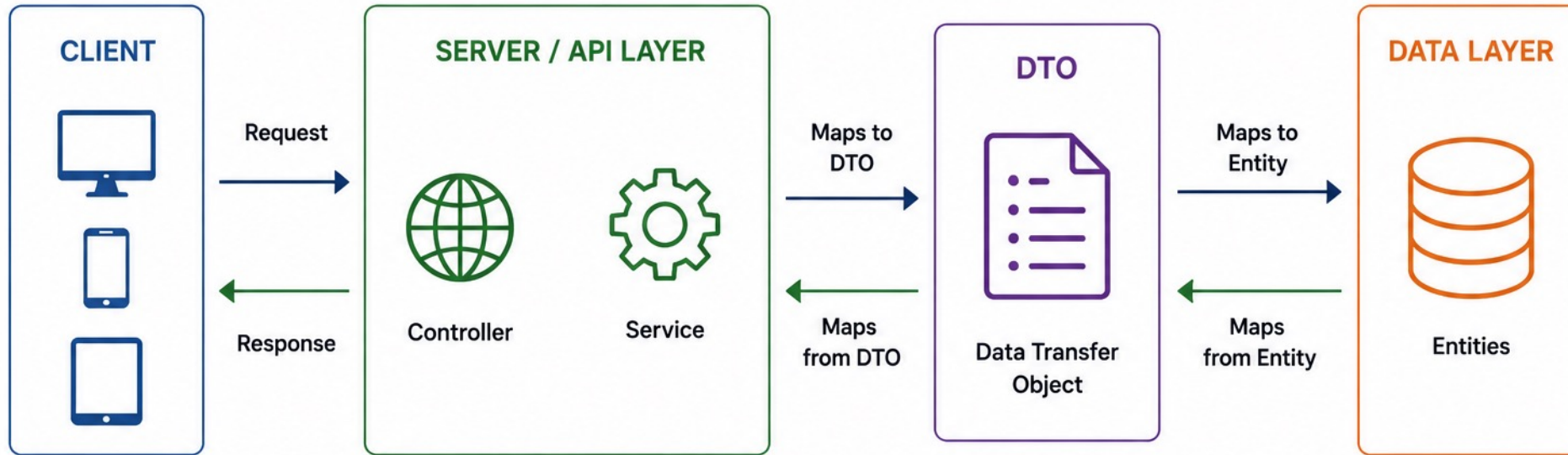


Microservices

Transfer only relevant data between services.

KEY TAKEAWAY Use DTOs to define clear contracts, protect your domain, and deliver the right data to the right consumer — every time.

DTOs (Data Transfer Objects)



WHY DTOs MATTER



Security

Hide internal data
and sensitive fields



Performance

Transfer only the
required data



Loose Coupling

Decouple API from
internal models



Flexibility

Evolve internal models
without breaking APIs



Maintainability

Simpler, cleaner and
easier to maintain

ENTITY VS DTO

Use Entities to model your business and store data. Use DTOs to communicate data safely and efficiently.

ASPECT	ENTITY	DTO
Purpose	Model the domain & persist data	Transfer data across boundaries
Scope	Internal (application)	External (API / clients)
Contains	Entities represent persisted domain state and may include fields or relationships that should not be exposed through an API.	Only required fields
Relationships	Yes (to other entities)	No, or simplified
Change Impact	High (affects many areas)	DTOs can evolve independently from entities, but public contract changes must still be managed for compatibility.

ENTITY VS DTO — EXAMPLES & BEST PRACTICES

Concrete Entity/DTO name pairs and guidance for choosing dedicated DTOs over exposing persistence entities directly.

EXAMPLES

- Entities: UserEntity, OrderEntity, ProductEntity → DTOs: UserResponseDto, CreateUserRequestDto, UserSummaryDto.

BEST PRACTICES

- For public, partner-facing, or long-lived APIs, use dedicated DTOs rather than exposing persistence entities directly. Convert between entities and DTOs through explicit mapping logic.

KEY TAKEAWAY Entities model your world. DTOs shape your communication — use the right tool in the right place.

Entity vs DTO

ENTITY	VS	DTO
 Represents domain data and business state	 Purpose	 Carries data between layers or over the network
 Used within the domain (Entity Layer)	 Scope	 Used across layers or external boundaries
 May contain business logic, rules, and behaviors	 Behavior	 No business logic, only data
 May have relationships with other entities	 Dependencies	 No references to other domain objects
 Persisted in the database (mapped by ORM)	 Persistence	 Not persisted, used for data transfer only
 Rich structure with many fields and relationships	 Structure	 Flat or simple structure with only required fields
 Used by Service/Repository layers for domain operations	 Use Case	 Used by Controller, API, or client layers
 Changes can impact domain logic and persistence	 Change Impact	 Changes are isolated, safe for external contracts

TRY IT YOURSELF — EXERCISE 1: ENTITY VS DTO

Reinforces: the Entity vs DTO purpose, scope, and contents you just saw.

STEP	WHAT TO DO
Goal	Explain why Northstar HTTP/SOAP payloads should not be persistence entities
File	notes/lab14-entity-vs-dto.md (in examples/module-14-exercises/)
Leak risks	Two leaks: internal flags, lazy relations, or audit columns in responses
Amina fields	customerId, fullName, status — no persistence annotations
Boundary	Pre-lab only — no Spring @Valid controllers yet
Reference	exercises/exercise-01-entity-vs-dto.md

```
// CustomerEntity (persistence) vs CustomerDto (API) for Amina Khan
class CustomerEntity { Long id; String fullName; CustomerStatus status; }
    boolean internalRiskFlag; Instant auditCreatedAt; // never leaves this class
record CustomerDto(String customerId, String fullName, String status) {}
// no persistence annotations on the DTO
```

TRY IT NOW Complete Exercise 1 before continuing — see exercises/exercise-01-entity-vs-dto.md.

REQUEST DTO DESIGN

A Request DTO captures the data sent by the client for a specific operation — validates input and protects your domain model.

PRINCIPLES

- Client-controlled fields only — the client supplies just what it's allowed to set.
- Validation-ready — annotate fields with Jakarta Validation constraints.
- Use separate create, replace, and patch contracts when their required fields or semantics differ.
- Over-posting protection — never bind server-controlled or internal fields from client input.

EXAMPLE REQUEST DTO

```
public record CreateCustomerRequest(  
    @NotBlank String fullName,  
    @NotBlank @Email String email  
) {}  
// Records suit immutable request/response DTOs;  
// check framework & serialization support.
```

REQUEST DTO DESIGN — WHAT TO EXCLUDE & PATTERNS

Fields client input must never set, and the standard shapes of create, replace, and patch requests.

WHAT TO EXCLUDE

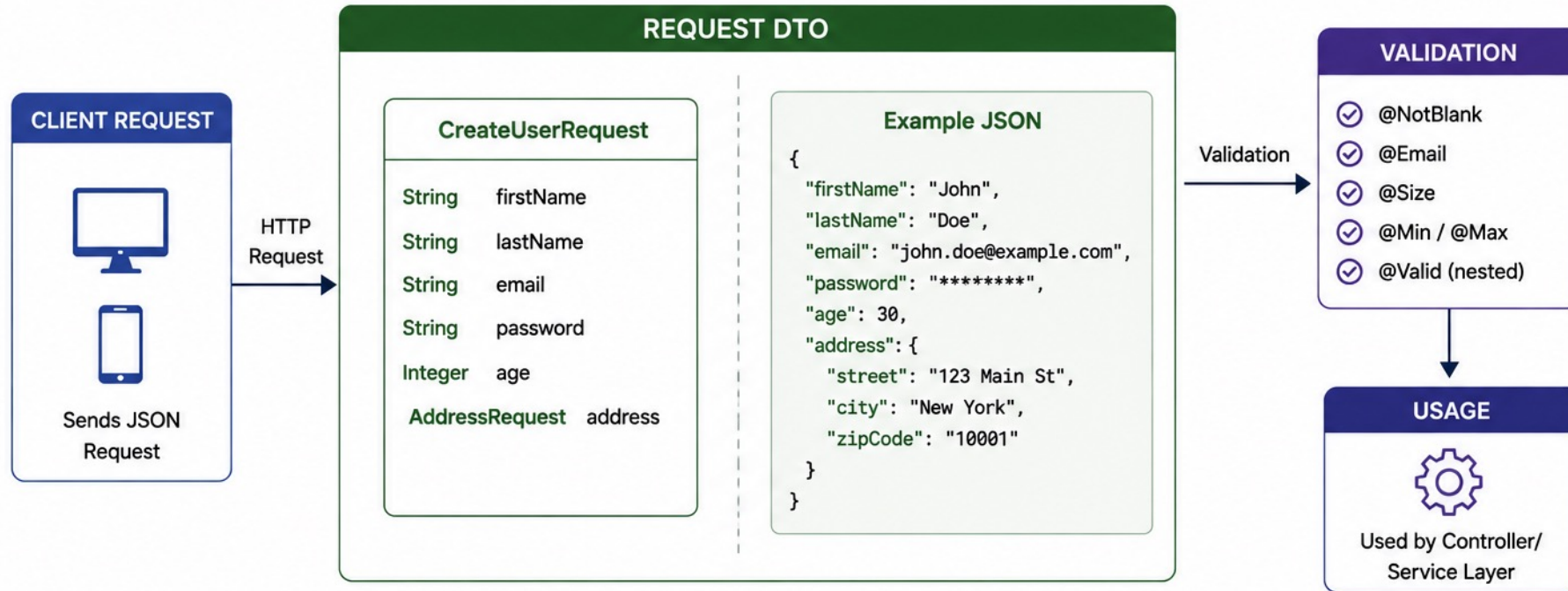
- Server-generated fields (id, createdAt, status); sensitive fields (internal flags, audit info) that the client should never set.

COMMON REQUEST PATTERNS

- Create Request (POST) — full resource creation. Replace Request (PUT) — full resource replacement. Patch Request (PATCH) — partial modification; null may mean absent, explicit clearing, or invalid depending on design. Use JSON Merge Patch, JSON Patch, explicit optional wrapper types, or field-presence tracking for patch — don't reuse a normal nullable-field DTO unless absence-vs-explicit-null is clearly handled.

KEY TAKEAWAY Design Request DTOs around client-controlled fields, validation, and clear create/replace/patch semantics.

Request DTO Design



BEST PRACTICES



1. SECURITY

Include only required fields. Never expose internal data.



2. VALIDATION

Validate all inputs using Bean Validation annotations.



3. LOOSE COUPLING

Keep DTOs separate from entities to avoid coupling and protect internal model.



4. SIMPLICITY

Keep DTOs focused, small and easy to understand.



5. EXTENSIBILITY

Design for future changes using optional fields and versioning.

RESPONSE DTO DESIGN

A Response DTO defines the data returned after a request is processed — consistent, secure, and efficient.

PRINCIPLES

- Server-controlled fields only — computed or persisted values the client cannot set.
- Consistent representation shape — same structure for the same resource type.
- Sensitive-data filtering — never expose passwords, internal flags, or stack traces.
- Pagination and links — include paging metadata and navigation links for collections.

COMMON PATTERNS

- Standard response, paged response (with pagination), summary/count response, action result response.

RESPONSE DATA ELEMENTS

- Also include metadata (pagination, timestamps, correlationId) and HATEOAS links when applicable. Avoid a redundant textual status field when the HTTP status code already communicates success. Additive fields are generally backward-compatible when consumers tolerate unknown properties. Confirm client behavior and published schema rules.

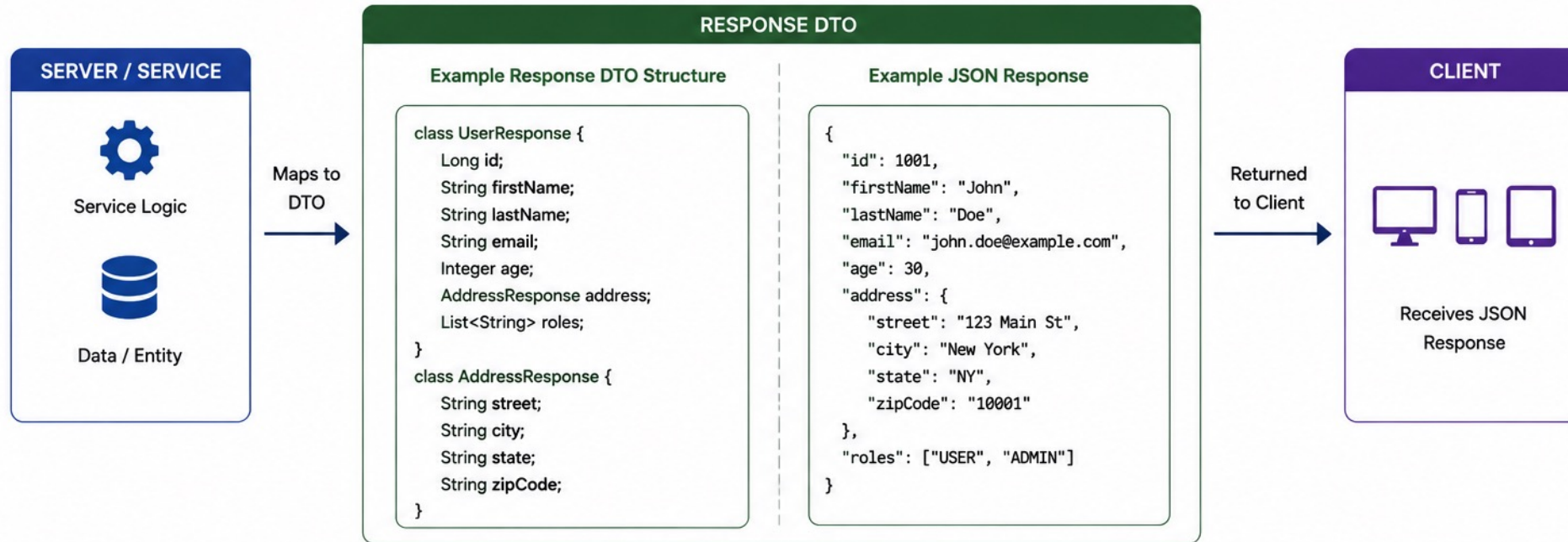
EXAMPLE SUCCESS RESPONSE

```
// Direct resource response
{ "id": 1, "name": "Jane" }

// Standard envelope (when required)
{ "data": { ... }, "meta": { ... } }
```

KEY TAKEAWAY A great API doesn't just process requests — it communicates effectively. Return the right data, in the right format.

Response DTO Design



DESIGN PRINCIPLES

- Expose Only Necessary Data**
Return only the fields required by the client.
- Hide Internal Details**
Do not expose sensitive or internal data.
- Loose Coupling**
DTOs decouple API contracts from domain models.
- Consistent Structure**
Use consistent naming and format across APIs.

BEST PRACTICES

- Use meaningful and consistent field names.
- Support pagination for large collections.
- Include metadata when useful.
- Use nested DTOs for related data.
- Version your API responses when needed.

COMMON RESPONSE ENVELOPE

```
{
  "success": true,
  "message": "Request successful",
  "data": {
    ... // Response DTO
  },
  "metadata": {
    "timestamp": "2025-05-21T10:15:30Z",
    "requestId": "a1b2c3d4-e5f6-7890"
  }
}
```

- Status of the request
- Message for the client
- Response DTO payload
- Additional metadata

MAPPING STRATEGIES

Convert Entities to DTOs (and back) using the right approach for your complexity, team, and performance needs.

STRATEGY	CHARACTERISTICS	SUITABLE FOR
Manual Mapping	Explicit, no extra framework, more boilerplate	Small mappings or complex custom logic
ModelMapper	Convention-based, runtime mapping	Prototypes or simple mappings
MapStruct	Compile-time generated, type-safe	Larger production codebases
Legacy reflection tools	Flexible but harder to reason about	Existing legacy systems

MAPPING STRATEGIES — BEST PRACTICES & FLOW

How to keep mapping logic maintainable, and the standard Entity-to-DTO-to-consumer flow.

BEST PRACTICES

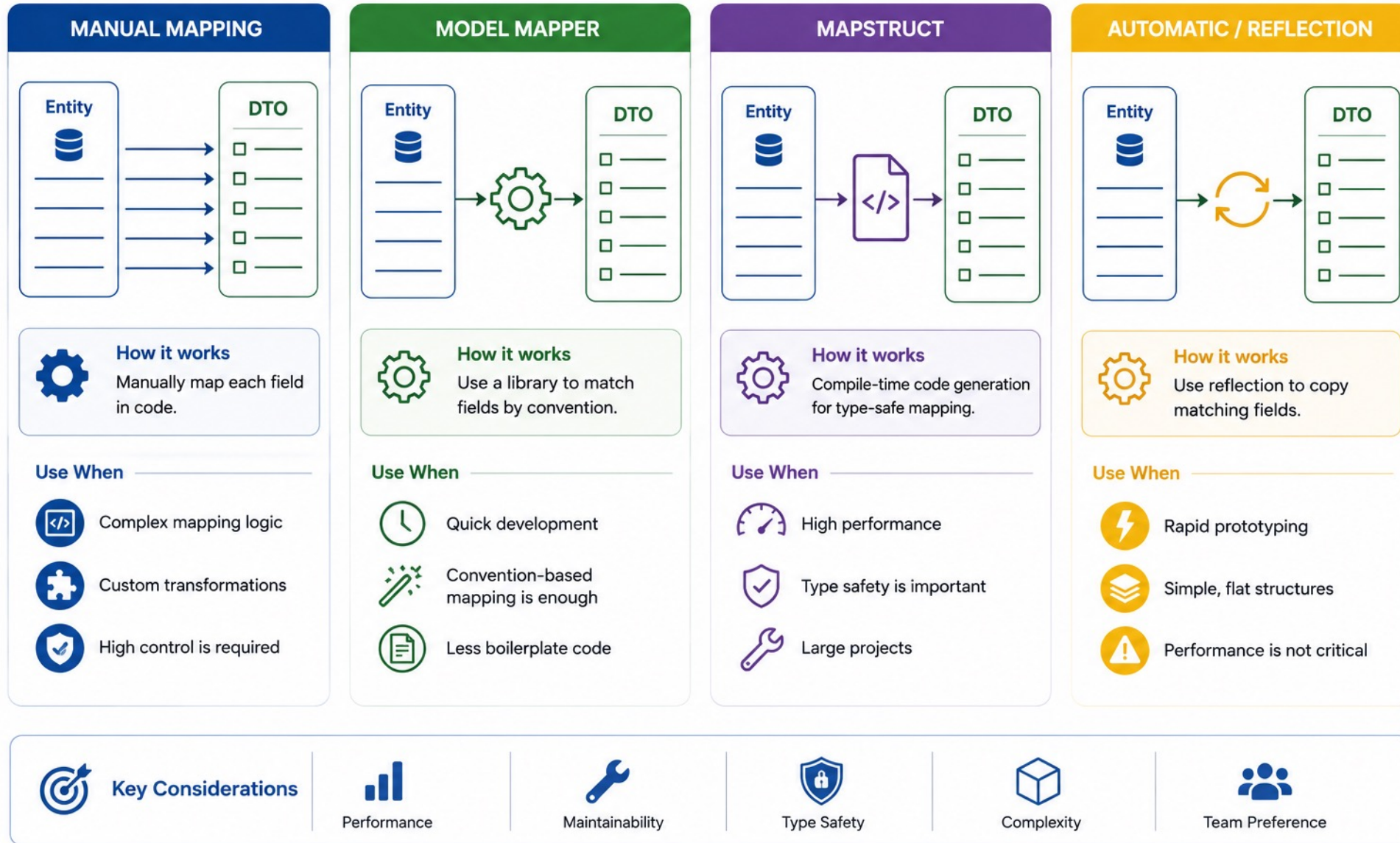
- Keep mapping logic in a dedicated layer; validate before mapping to Entities.
- MapStruct is a strong choice for repetitive, type-safe mappings. Use manual mapping when transformation rules are small, security-sensitive, or highly domain-specific.
- Handle nulls and collections carefully; write unit tests for custom mappings.

MAPPING FLOW

- Entity → Mapping Layer → DTO → API Consumer. Treat request-to-domain and domain-to-response mapping as separate operations with separate security and business rules — clients cannot set IDs, audit fields are server-generated, existing entity state is preserved, patch semantics need field-presence handling, and relationships may be resolved through services.

KEY TAKEAWAY Choose the mapping strategy that balances productivity, maintainability, performance, and team expertise.

Mapping Strategies



MAPPING STRATEGIES — CODE EXAMPLE

Two ways to convert between Entities and DTOs: explicit manual mapping, and compile-time generated MapStruct.

EXAMPLE: MANUAL MAPPING VS MAPSTRUCT

```
// Manual mapping — explicit, no extra dependency
public CustomerDto toDto(CustomerEntity entity) {
    return new CustomerDto(
        entity.getId(),
        entity.getFullName(),
        entity.getStatus().name());
}

// MapStruct — compile-time generated, type-safe
@Mapper(componentModel = "spring")
public interface CustomerMapper {

    @Mapping(target = "status", expression = "java(entity.getStatus().name())")
    CustomerDto toDto(CustomerEntity entity);
}
```

- Keep the conversion in a dedicated mapper class — never inline it in a controller.

WHEN TO USE EACH

- Manual mapping — no extra dependency, full control; best for small DTOs or conversions with custom logic (enum-to-String, computed fields).
- MapStruct — an @Mapper interface is implemented for you at compile time; best for larger codebases with many repetitive, type-safe mappings.

KEY TAKEAWAY Manual mapping gives full control for small or custom conversions; MapStruct scales better for large, repetitive, type-safe mappings.

TRY IT YOURSELF — EXERCISE 2: MAPPER NO-LEAK RULE

Reinforces: the mapping strategies and Entity→DTO flow you just saw.

STEP	WHAT TO DO
Goal	Sketch toDto/toEntity rules that keep internals out of API responses
File	notes/lab14-mapper-rules.md (in examples/module-14-exercises/)
toDto (CUS-1001)	Map only id, fullName, status — nothing else
Forbidden	Password hashes, internal risk scores, raw SQL ids if different from public id
Activate DTO	Carries customerId only (+ correlation header outside the body)
Reference	exercises/exercise-02-mapper-no-leak.md

```
// CustomerMapper -- no-leak rule for CUS-1001
CustomerDto toDto(CustomerEntity e) {
    return new CustomerDto(e.getCustomerId(), e.getFullName(), e.getStatus().name());
}
// Never map passwordHash/riskScore -- DTOs before deep rules (Lab 15 owns transitions)
```

TRY IT NOW Complete Exercise 2 before continuing — see exercises/exercise-02-mapper-no-leak.md.

JAKARTA VALIDATION OVERVIEW

Validate early. Fail fast. Jakarta Validation, formerly Bean Validation and standardized in earlier versions through JSR 380, validates Java objects using annotations.



KEY TAKEAWAY Jakarta Validation is declarative, centralized, and consistent — apply it at the API boundary before business logic runs.

JAKARTA VALIDATION OVERVIEW — CONSTRAINTS & TARGETS

The built-in constraint annotations, and where structural validation applies.

COMMON BUILT-IN CONSTRAINTS

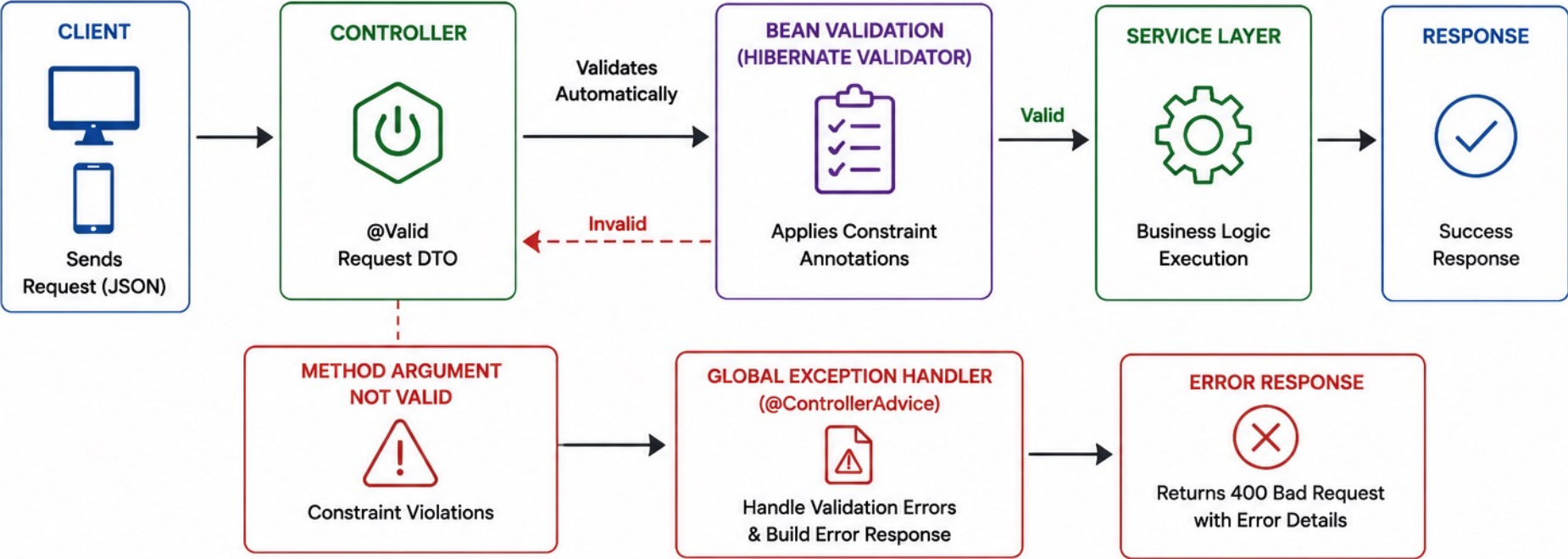
ANNOTATION	DESCRIPTION
<code>@NotNull</code> / <code>@NotBlank</code> / <code>@NotEmpty</code>	Value/String/Collection must not be null (or blank/empty)
<code>@Size(min, max)</code>	Size or length must be within range
<code>@Email</code> / <code>@Pattern(regex)</code>	Must be a valid email / match the given regex

VALIDATION TARGETS & TYPES

- Validates fields, method parameters, return values, nested objects (`@Valid`), and class-level cross-field rules (`@AssertTrue`).
- Use Jakarta Validation for structural and local cross-field constraints. Enforce business rules in the domain or service layer where business context is available.

KEY TAKEAWAY Jakarta Validation is declarative, centralized, and consistent — apply it at the API boundary before business logic runs.

Bean Validation



COMMON CONSTRAINT ANNOTATIONS	
@NotNull	must not be null
@NotBlank	must not be null or blank (String)
@NotEmpty	must not be null or empty (Collection, Map, Array, String)
@Size(min, max)	size must be between min and max
@Min(value)	must be greater than or equal to value
@Max(value)	must be less than or equal to value
@Email	must be a well-formed email address
@Pattern(regexp)	must match the given regular expression

VALIDATION TARGETS

- Request DTO Fields
- Method Parameters
- Method Return Values
- Nested Objects (@Valid)

KEY BENEFITS

- Ensures data integrity
- Automatic validation
- Reduces boilerplate code
- Centralized error handling
- Improves API reliability

PRESENCE CONSTRAINTS

Ensure required data is present — @NotNull, @NotEmpty, and @NotBlank guard against missing, empty, or blank values.

EXAMPLE

```
public record CreateUserRequest(  
    @NotBlank String name  
) {}  
  
@NotNull  
@Valid  
private AddressRequest address;
```

WHERE CAN YOU USE IT?

- Entity/DTO fields, constructor parameters; controller or service parameters (needs framework support).
- Invalid: null, "" (empty), " " (whitespace only).
- Valid: any string with at least one non-whitespace character, e.g. "John Doe".
- @NotNull requires presence; @Valid validates the nested object — they are independent concerns.

PRESENCE CONSTRAINTS — @NOTNULL VS @NOTBLANK VS @NOTEMPTY

A side-by-side comparison of the three presence-checking annotations and what each one allows.

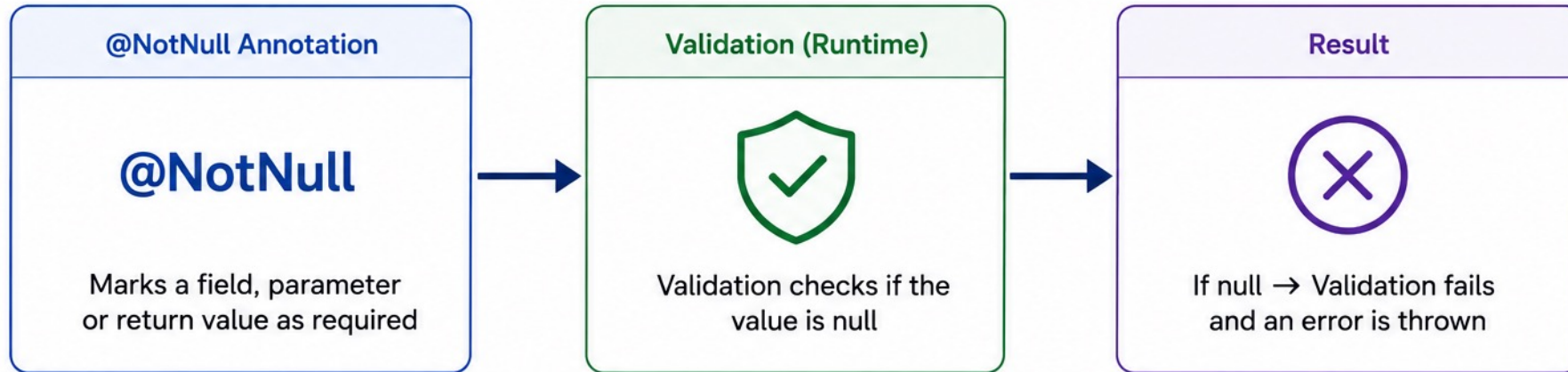
@NOTNULL VS @NOTBLANK VS @NOTEMPTY

ANNOTATION	APPLIES TO	ALLOWS EMPTY?	ALLOWS WHITESPACE?
@NotNull	Any type	Yes	Yes
@NotEmpty	CharSequence, collections, maps, arrays	No	Yes
@NotBlank	String	No	No

- @NotNull requires presence; @Valid validates the nested object — they are independent concerns (see the combined example above).

KEY TAKEAWAY @NotNull, @NotEmpty, and @NotBlank ensure required data is present — combine with @Valid for nested objects.

@NotNull Annotation

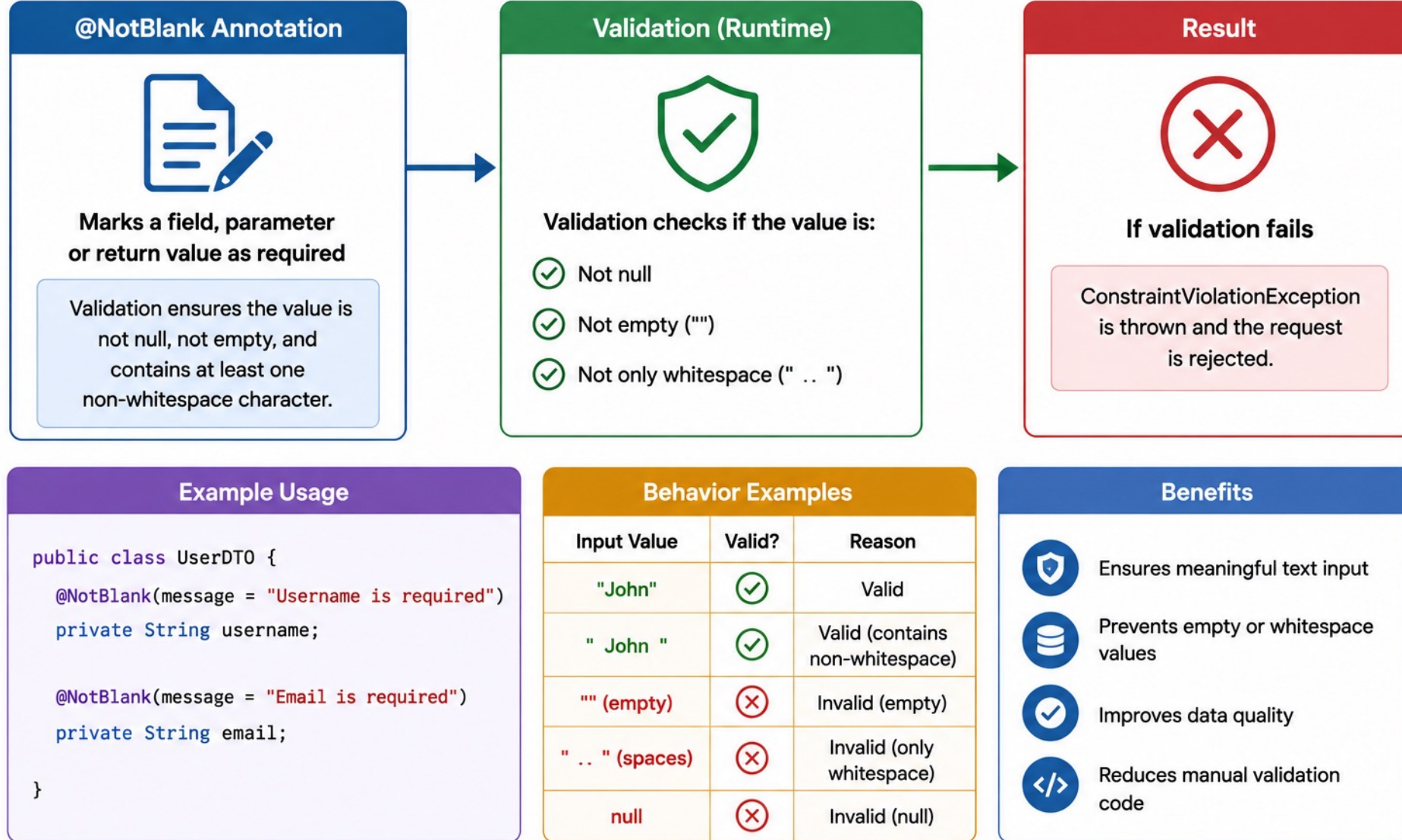


Example Usage

```
public class UserDTO {  
    @NotNull  
    private String username;  
  
    @NotNull  
    private String email;  
}
```

- Effect**
- ✓ Prevents null values
 - ✓ Ensures required data
 - ✓ Improves data integrity

@NotBlank Annotation



FORMAT AND SIZE CONSTRAINTS

Validate format and size — @Email, @Pattern, @Size, and @Min/@Max/@Positive keep data well-formed and within range.

EXAMPLE

```
@NotBlank
@Size(min = 3, max = 50)
private String name;

>Email
@NotBlank
private String email;
```

VALID VS INVALID

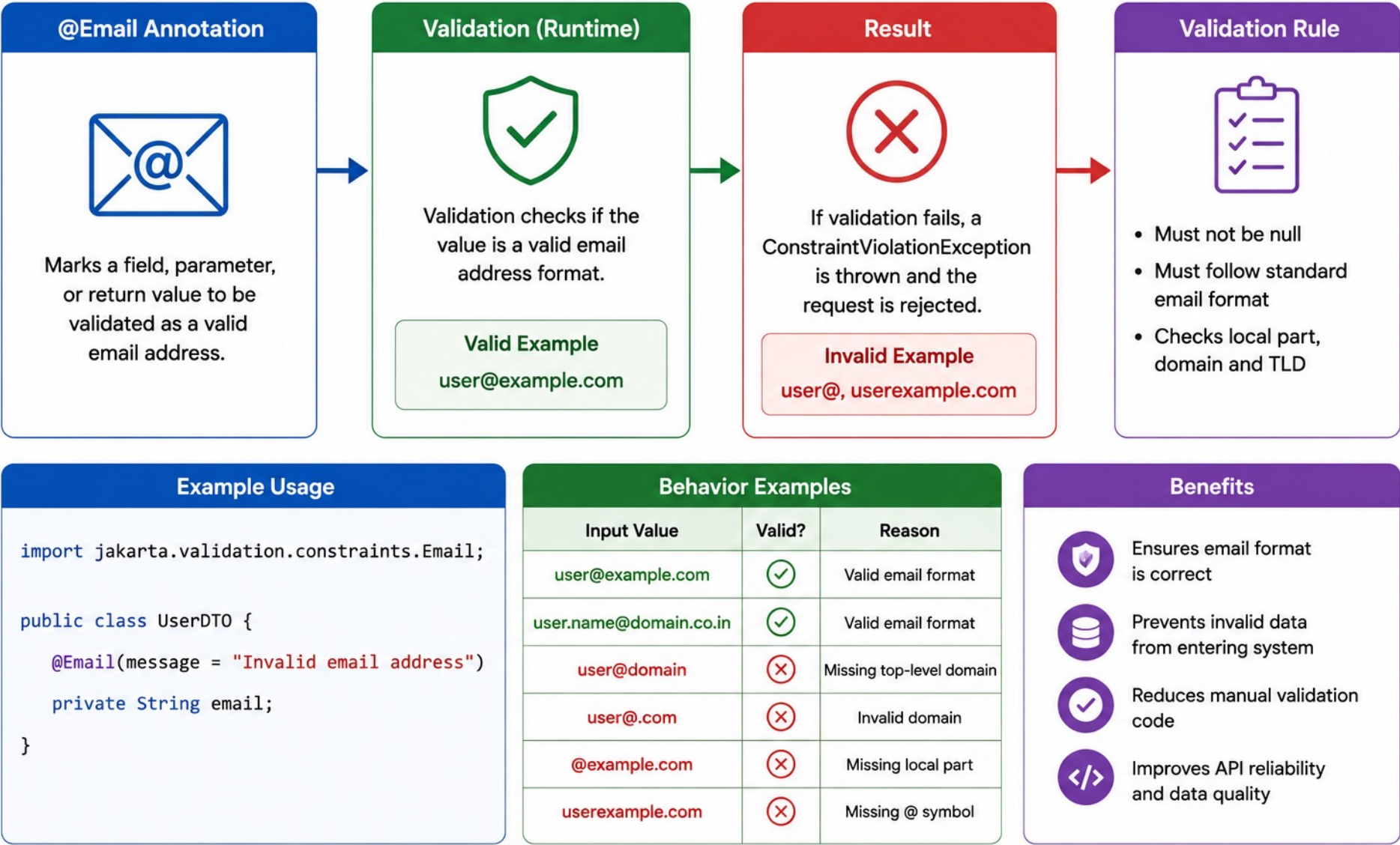
- Valid: john.doe@example.com, jane_doe123@domain.co.uk
- Invalid: john.doe@, john.doe.com, john..doe@example.com

BEST PRACTICES

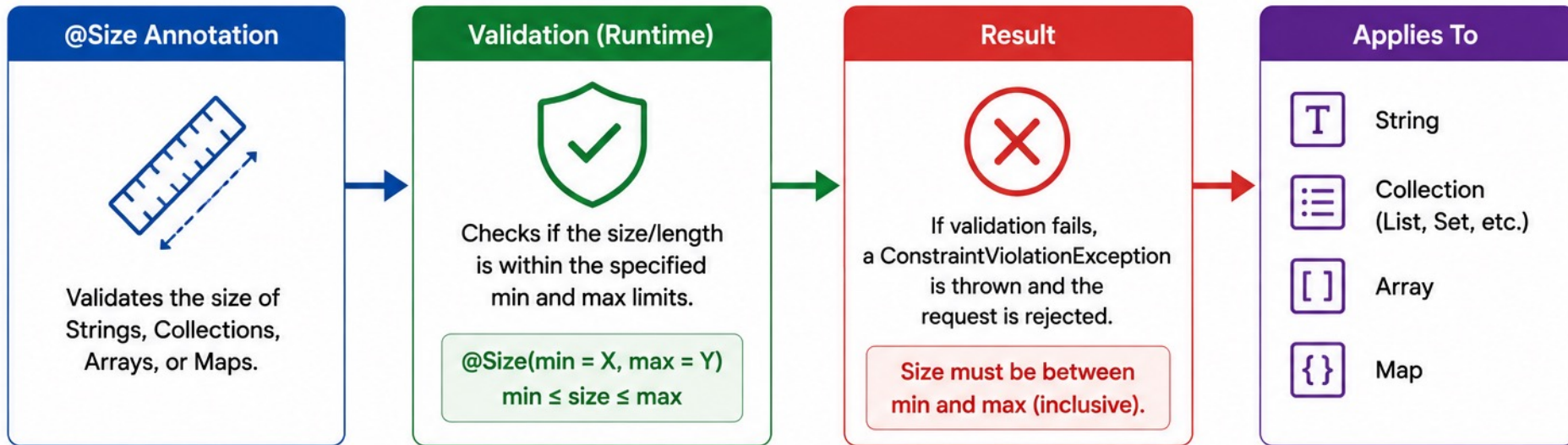
- @Size applies to String length, Collection/Array element counts, and Map entries; @Pattern(regex) matches a custom regular expression; @Min, @Max, and @Positive validate numeric ranges.
- Most format and size constraints treat null separately. Add a presence constraint when the value is required.
- @Email performs a basic syntax check. Verification links or other workflows are required to confirm ownership or deliverability.
- Combine @Email with a presence constraint when the field is required; always validate at the API boundary using @Valid.

KEY TAKEAWAY Format and size constraints keep data well-formed and within range — combine with a presence constraint when the value is required.

@Email Annotation



@Size Annotation



Example Usage

```
import jakarta.validation.constraints.Size;

public class UserDTO {
    @Size(min = 3, max = 20,
        message = "Name must be between 3 and 20 characters")
    private String name;

    @Size(min = 1, max = 5)
    private List<String> roles;
}
```

Behavior Examples

Input Value	@Size(min = 2, max = 5)	Result
"Hi" (length = 2)	✓	Valid
"Hello" (length = 5)	✓	Valid
"H" (length = 1)	✗	Invalid (too short)
"Welcome" (length = 7)	✗	Invalid (too long)
[A, B, C] (size = 3)	✓	Valid
[] (size = 0)	✗	Invalid (too short)

Key Benefits

- Ensures data size/length is within allowed limits
- Prevents invalid or excessive data
- Reduces manual validation code
- Improves API reliability and data quality

TRY IT YOURSELF — EXERCISE 3: ANNOTATE THE PAPER DTO

Reinforces: the presence, format, and size constraints you just saw.

STEP	WHAT TO DO
Goal	Mark required/optional constraints on a paper CreateCustomerRequest
File	notes/lab14-paper-dto.md (in examples/module-14-exercises/)
fullName	required, non-blank → @NotBlank
status	optional on create; defaults to PROSPECT (Ravi starts PROSPECT)
customerId	server-assigned or pattern CUS-#### → @Pattern
Reference	exercises/exercise-03-annotate-paper-dto.md

```
// Paper DTO -- documentation only, no Spring @Valid wired yet
record CreateCustomerRequest(
    @NotBlank String fullName, String status, // status optional, defaults PROSPECT
    @Pattern(regexp = "CUS-####") String customerId // server-assigned
) {} // correlation lab-request-001 stays in headers/logs, not a field
```

TRY IT NOW Complete Exercise 3 before continuing — see exercises/exercise-03-annotate-paper-dto.md.

VALIDATION ERROR HANDLING

When validation fails, return a standard, machine-readable error response with stable codes and a traceable correlation ID.

EXAMPLE ERROR RESPONSE

```
{
  "type": "https://api.northstar.example/problems/validation",
  "title": "Validation failed",
  "status": 400,
  "detail": "One or more fields are invalid.",
  "instance": "/api/v1/customers",
  "errors": [
    { "field": "email", "message": "must be a well-formed email address" }
  ],
  "correlationId": "lab-request-001"
}
```

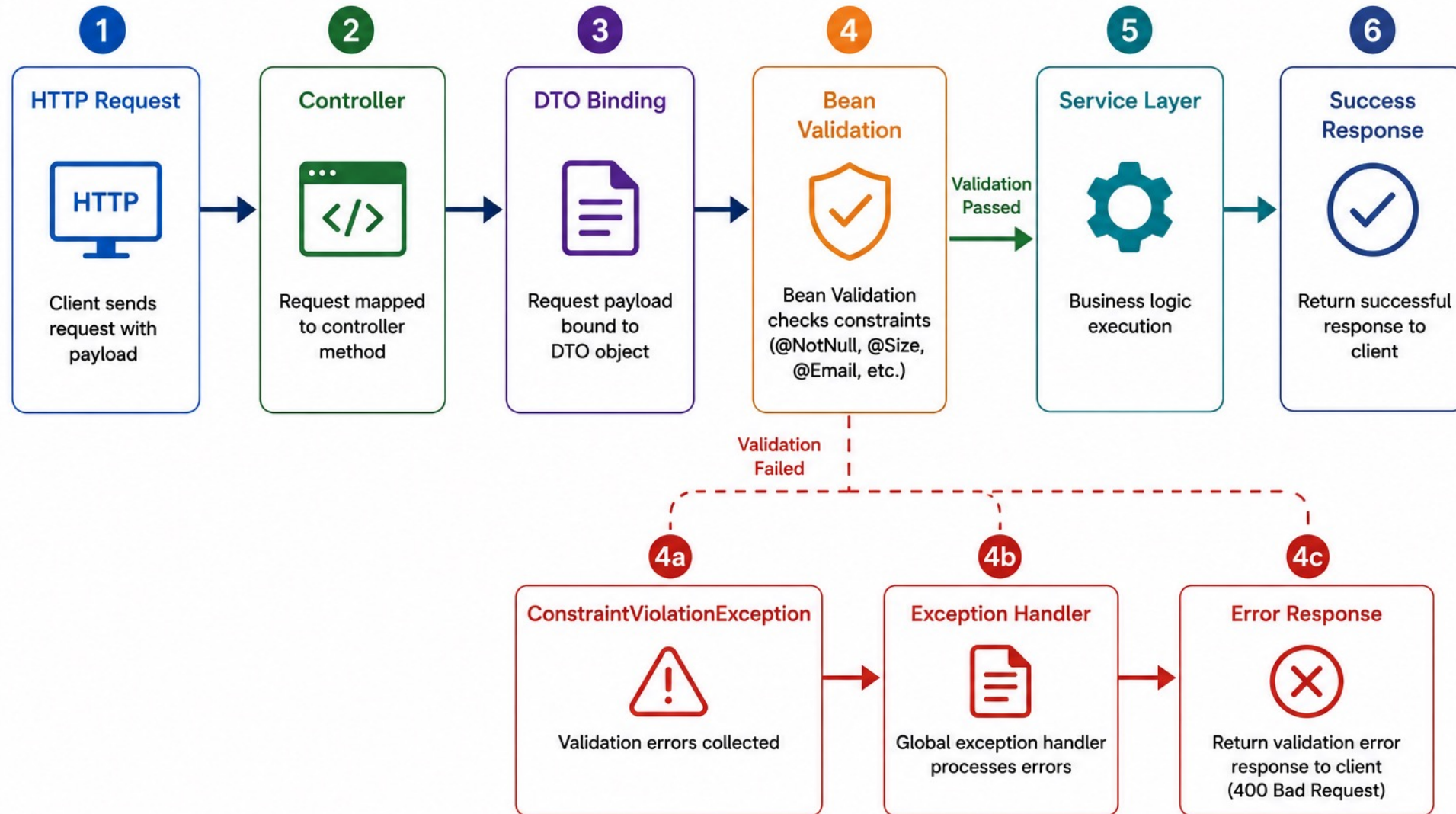
- "errors" and "correlationId" are extensions — non-standard fields added on top of the RFC 9457 base.

STABLE ERROR CODES & CORRELATION ID

- Define a stable external error code, consumer-safe message, field path, correlation ID, and optional localized text — never expose raw internal validation messages, e.g. { "code": "CUSTOMER_EMAIL_INVALID", "field": "email", "message": "Enter a valid email address." }
- Correlation ID comes from an incoming request header or is generated at the boundary; include it in logs, error responses, and service calls — never use it as a security token.

KEY TAKEAWAY Return RFC 9457 problem details with stable error codes and a correlation ID so every rejection is traceable and safe to show clients.

Validation Workflow



TRY IT YOURSELF — EXERCISE 4: INVALID CASES CATALOG

Reinforces: the validation error handling and RFC 9457 response you just saw.

STEP	WHAT TO DO
Goal	Catalog invalid requests you will assert later in Lab 14
File	notes/lab14-invalid-cases.md (in examples/module-14-exercises/)
Create invalids	Blank name; status = "ACTVE" typo; oversized name
Activate invalids	Activate missing id; activate CUS-9999 unknown
Valid control	Create Ravi-shaped PROSPECT with a non-blank name
Reference	exercises/exercise-04-invalid-cases.md

```
// Invalid create payloads to assert in Lab 14
{ "fullName": "", "status": "PROSPECT" } // blank name
{ "fullName": "Amina Khan", "status": "ACTVE" } // status typo
{ "fullName": "A...A" } // 300 chars, oversized name
// Valid control: { "fullName": "Ravi Shah", "status": "PROSPECT" }
```

TRY IT NOW Complete Exercise 4 before continuing — see exercises/exercise-04-invalid-cases.md.

HTTP API CONTRACT DESIGN

An API contract is a formal agreement between provider and consumer. This section focuses on REST-style HTTP contracts.

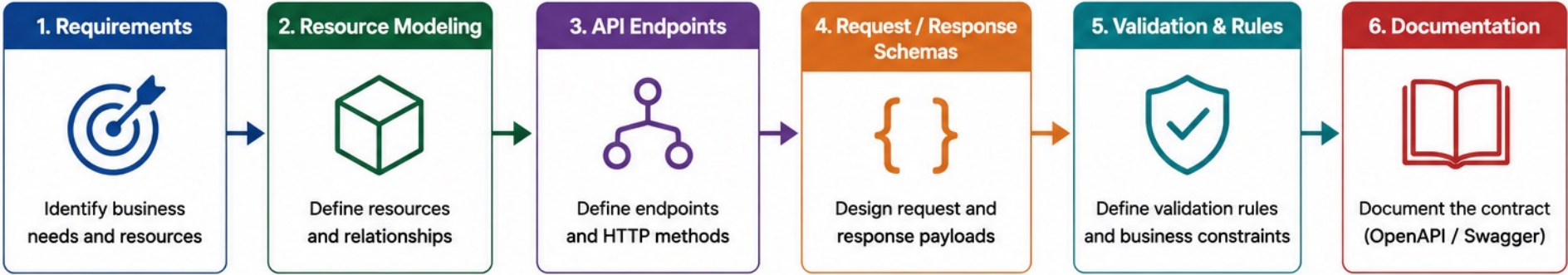
COMPONENT	DEFINES	EXAMPLE
Endpoints	HTTP method, path, query parameters	POST /api/v1/users
Request Contract	Request DTO structure, validation rules	name, email, age fields
Response Contract	Response DTO structure, pagination	201 Created + UserResponseDto
Error Contract	Standard error structure, codes	400 Bad Request + field errors
Versioning	URI or header versioning strategy	/api/v1/users

BEST PRACTICES

- Keep contracts simple and consistent; use nouns for resources and standard HTTP methods; document with OpenAPI/Swagger.
- Specify content types, authentication/authorization expectations, and idempotency for unsafe methods.
- Define pagination, rate limits, field semantics, and nullability for every resource.
- Provide request/response examples, a deprecation policy, and clear compatibility guarantees.

KEY TAKEAWAY A clear API contract ensures consistency, reduces integration issues, and speeds up development.

API Contract Design



API Contract Components

Base URL
API base endpoint

Endpoints
Resource paths and operations

HTTP Methods
GET, POST, PUT, PATCH, DELETE

Headers
Authentication, Content-Type, etc.

Request Schema
Input data structure

Response Schema
Output data structure

Error Responses
Standard error codes and messages

Example: User Resource Contract

Endpoint

GET /api/v1/users/{id}

Description: Retrieve user by ID

Request

Headers:

- Authorization: Bearer <token>
- Accept: application/json

Path Parameters:

Name	Type	Required	Description
id	string	Yes	User ID

Response (200 OK)

Content-Type: application/json

```
{  "id": "123",  "name": "John Doe",  "email": "john@example.com",  "status": "active",  "createdAt": "2024-06-01T10:00:00Z"}
```

Error Responses

Status Code	Description	Example
400	Bad Request	{ "error": "Invalid input" }
401	Unauthorized	{ "error": "Unauthorized" }
404	Not Found	{ "error": "User not found" }
500	Internal Server Error	{ "error": "Server error" }

Best Practices

Versioning
Use URI versioning (e.g., /api/v1/)

Consistent Naming
Use clear and consistent naming conventions

Filtering & Pagination
Support query params for filtering, sorting, pagination

Security
Use authentication and authorization

Documentation
Keep API docs up to date (OpenAPI / Swagger)

API CONTRACT DESIGN — CODE EXAMPLE

A Spring REST controller implementing the User resource contract — typed parameters in, a DTO and an explicit status code out.

EXAMPLE CONTROLLER

```
@RestController
@RequestMapping("/api/v1/users")
public class UserController {

    @GetMapping("/{id}")
    public ResponseEntity<UserResponse> getUser(
        @PathVariable Long id) {
        UserResponse body = userService.findById(id);
        return ResponseEntity.ok(body);
    }
}
```

BEST PRACTICES

- Keep controllers thin — delegate lookups and business rules to the service layer.
- Return a DTO from the controller, never the entity, so the response shape stays stable.
- Use ResponseEntity when the endpoint needs to control status codes or headers explicitly.
- Match the path, method, and payload exactly to the published contract — consumers depend on it.

REQUEST / RESPONSE MAPPING

- @PathVariable binds {id} from the URI to the id method parameter.
- ResponseEntity<UserResponse> lets the controller return the DTO body with an explicit HTTP status (200 OK).

KEY TAKEAWAY A REST controller should implement the contract exactly — typed parameters in, a DTO and an explicit status code out.

VERSIONING API CONTRACTS

Evolve APIs without breaking consumers — choose a strategy, apply a documented compatibility policy, and keep a clear lifecycle.

STRATEGY	EXAMPLE	TRADE-OFF
URI Path	/api/v1/users	Easy to understand, cache-friendly
Query Parameter	/api/users?version=1	Easy to add, but may complicate routing, caching, documentation, and client consistency.
Custom Header	X-API-Version: 1	Less visible and harder to discover; requires explicit client headers; can complicate caching and observability.
Media Type	Accept: application/vnd.acme.user+json;v=1	Flexible, more complex to implement

VERSIONING API CONTRACTS — SEMANTIC VERSIONING

Apply a documented compatibility policy and reserve major versions for incompatible changes.

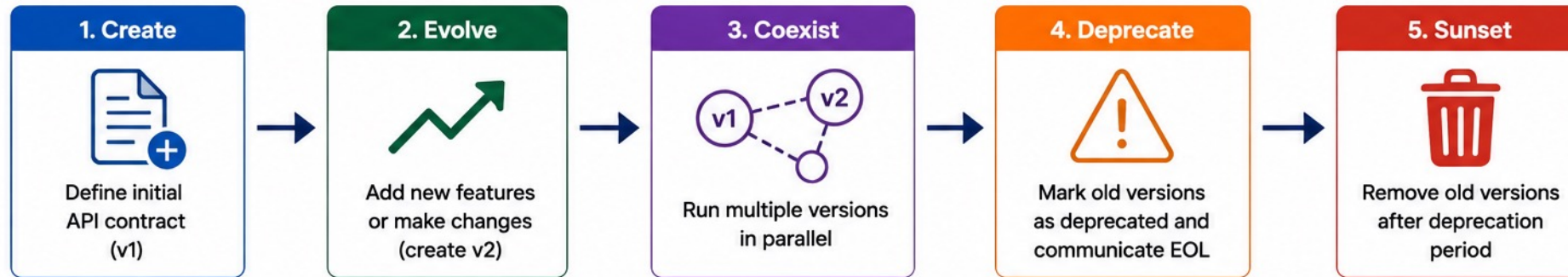
SEMANTIC VERSIONING

- Use a documented compatibility policy. Public HTTP APIs commonly expose only major versions, while internal releases may still follow semantic versioning.
- Deprecate, don't delete immediately — set clear sunset dates and provide migration guides.
- Do not create a new version for every additive change. Reserve new major versions for incompatible contract changes. Additive optional fields, new endpoints, new enum values, changed validation constraints, and changed error behavior are usually not version-triggering — though some "additive-looking" changes can still break strict clients.

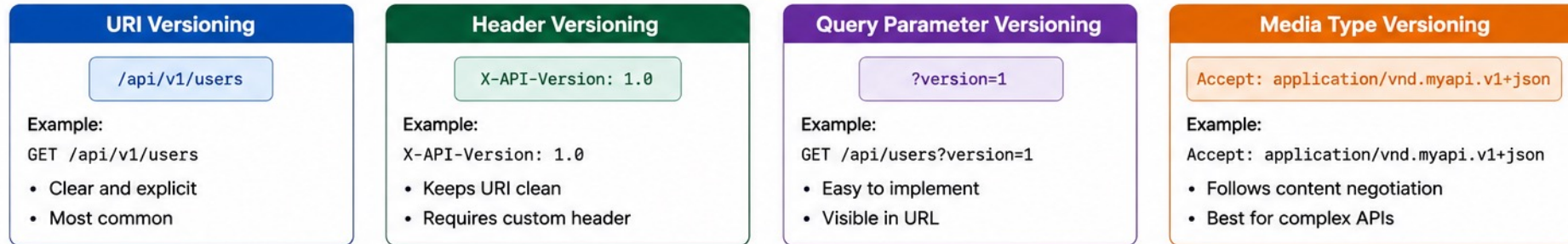
KEY TAKEAWAY Versioning is about managing change, not just changing the URL — communicate clearly and evolve safely.

Versioning API Contracts

API Version Lifecycle



Versioning Strategies



Versioning Comparison				
Strategy	Example	Pros	Cons	Best Use Case
URI Versioning	<code>/api/v1/users</code>	Simple, clear, easy to cache	URI changes, may clutter paths	Public APIs, most common choice
Header Versioning	<code>X-API-Version: 1.0</code>	Clean URI, flexible	Less visible, harder to test	Internal APIs, advanced clients
Query Parameter	<code>?version=1</code>	Easy to implement, visible	Affects caching, less RESTful	Simple APIs, quick adoption
Media Type Versioning	<code>application/vnd.myapi.v1+json</code>	RESTful, uses content negotiation	Complex to implement, not widely used	Enterprise APIs, content-aware clients

Best Practices	
	Use semantic versioning (v1, v2, v3...)
	Follow deprecation policy and timelines
	Communicate changes clearly
	Ensure backward compatibility when possible
	Document each version separately
	Monitor usage and plan version sunset

VERSIONING API CONTRACTS — CODE EXAMPLE

URI versioning in Spring: v1 and v2 controllers coexist, each returning its own response DTO.

EXAMPLE: URI VERSIONING

```
@RestController
@RequestMapping("/api/v1/users")
class UserControllerV1 {

    @GetMapping("/{id}")
    UserResponseV1 getUser(@PathVariable Long id) {
        return mapper.toV1(service.findById(id));
    }
}
// v2 adds a field – same resource, richer response
record UserResponseV2(Long id, String email, Instant createdAt) {}
```

BEST PRACTICES

- Reserve a new major version for breaking changes only — added optional fields are not breaking.
- Keep old versions running until consumers migrate; publish a deprecation and sunset date.
- Name versioned types clearly (UserResponseV1, UserResponseV2) so intent is obvious in code.
- Pick one versioning strategy — URI, header, or media type — and apply it consistently.

KEY TAKEAWAY Versioning is a code decision too — coexisting controllers and typed response DTOs let v1 and v2 evolve independently.

WHY THIS WORKS

- Each major version gets its own controller and response type — v1 keeps working unchanged.
- Additive fields, like createdAt on v2, do not require a new version at all.

ENTERPRISE VALIDATION BEST PRACTICES (1 OF 2)

Validation is the first line of defense against bad data, security issues, and system failures.



Validate Early & Consistently

- At the API boundary, before business logic runs.



Defense in Depth

- Gateway (size, protocol, abuse controls); API boundary (structural DTO validation); Service/domain (business rules); Database (integrity constraints).



Comprehensive Constraints

- Null/empty, format, range, custom rules.

ENTERPRISE VALIDATION BEST PRACTICES (2 OF 2)

Validation is the first line of defense against bad data, security issues, and system failures.



Secure Your APIs

- Validate formats; encode or parameterize at the point of use (SQL, HTML, URLs, file paths).



Support i18n

- Externalize messages for multiple languages.



Test Your Validations

- Unit, negative, contract, and edge-case tests.

KEY TAKEAWAY Great validation leads to better data, happier consumers, fewer defects, and more reliable enterprise systems.

TRY IT YOURSELF — EXERCISE 5: VALIDATOR TODOS (STARTER)

Reinforces: the programmatic validation bootstrap you just saw — fill-in-the-blank, not from scratch.

STEP	WHAT TO DO
Goal	FILL-IN-THE-BLANK STARTER — complete the ValidatorFactory checklist (no Spring @Valid)
File	notes/lab14-validator-todos.md (in examples/module-14-exercises/)
Fill in	Bootstrap blanks: buildDefaultValidatorFactory(), getValidator(), violation counts
Invalid cases	Add: blank fullName; unknown status; null customerId on activate
Self-check	Confirm Spring @Valid blank is answered "no" for this pre-lab
Reference	exercises/exercise-05-fill-validatorfactory-todos.md

```
ValidatorFactory factory = ____;  Validator validator = ____;
Invalid blank name -> expect ____ violations
Invalid status TYPO -> expect ____
Valid Amina ACTIVE sketch -> expect ____ violations
Spring @Valid in this pre-lab? ____
```

TRY IT NOW Complete Exercise 5 (STARTER) — see exercises/exercise-05-fill-validatorfactory-todos.md.

TRY IT YOURSELF — EXERCISE 6: LAB 14 PREP CHECKLIST

Reinforces: everything from Exercises 1-5 — confirm you're ready for Lab 14.

STEP	WHAT TO DO
Goal	Confirm DTO/validation prep is complete without doing Lab 14 itself
File	notes/lab14-prep-checklist.md (in examples/module-14-exercises/)
Artifacts	Entity/DTO note, paper DTO, mapper rules, and validator TODOs all exist
Boundary	DTOs before deep service rules; no Spring @Valid live in this prep
Fixtures	Recall the two fixtures: Amina ACTIVE and Ravi PROSPECT
Reference	exercises/exercise-06-lab14-prep-checklist.md

```
[ ] notes/lab14-entity-vs-dto.md           // Exercise 1
[ ] notes/lab14-paper-dto.md               // Exercise 2
[ ] notes/lab14-mapper-rules.md            // Exercise 3
[ ] notes/lab14-validator-todos.md         // Exercise 4
[ ] notes/lab14-invalid-cases.md           // Exercise 4 -- pass requires this one
```

TRY IT NOW Complete Exercise 6 before continuing — see exercises/exercise-06-lab14-prep-checklist.md.

LAB OVERVIEW – DTOS AND VALIDATION

Design Customer DTOs and apply Jakarta Bean Validation programmatically for the Northstar CRM contract boundary.

WHAT YOU WILL LEARN

- Design CustomerRequestDTO and CustomerResponseDTO for create/read flows.
- Use common validation annotations (@NotNull, @NotBlank, @Email, @Size).
- Trigger validation programmatically and map entity to DTO without leaking fields.

PREREQUISITES

- JDK 21 and Maven 3.9+ (no Spring Boot in this lab).
- jakarta.validation-api, Hibernate Validator, and a compatible Jakarta Expression Language implementation if required by the selected configuration (e.g. org.glassfish:jakarta.el).
- Working CRM project (Customer, CustomerStatus) from Labs 9-12.

LAB OVERVIEW – HOW IT WORKS & VALIDATION BOOTSTRAP

The four-step validation flow, and the programmatic bootstrap code you'll use in Lab 14.

HOW IT WORKS



PROGRAMMATIC VALIDATION BOOTSTRAP

```
ValidatorFactory factory = Validation.buildDefaultValidatorFactory();
Validator validator = factory.getValidator();
Set<ConstraintViolation<CustomerRequestDto>> violations = validator.validate(request);
try (ValidatorFactory factory = Validation.buildDefaultValidatorFactory()) { Validator validator =
factory.getValidator(); }
```

KEY TAKEAWAY Always validate before mapping to the entity, and keep the correlation ID visible on failures.

LAB TASKS AND DELIVERABLES (1 OF 2)

Design Customer DTOs, validate programmatically, map safely, and prove rejection of invalid payloads.

#	TASK	DETAILS	FORMAT
1	Project Setup	Copy lab12-crm to lab14-crm; add Jakarta Validation deps.	n/a
2	Design DTOs	CustomerRequestDTO (validated) + CustomerResponseDTO.	.java
3	Implement CustomerMapper	Map Customer to/from DTOs; never leak entity fields.	.java

LAB TASKS AND DELIVERABLES (2 OF 2)

Validation, testing, and documentation — tasks 4-6, plus the success criteria for Lab 14.

#	TASK	DETAILS	FORMAT
4	Validate in CustomerApiFacade	Validator.validate() before service.createCustomer().	.java
5	Prove Rejection with Tests	JUnit tests: invalid email, blank name, valid Amina Khan.	.java
6	Document the Contract	README validation table; correlation ID lab-request-001.	.md

SUCCESS CRITERIA

- Valid CustomerRequestDTO maps and saves via the facade; invalid payloads never reach CustomerService; tests are green.
- Mapping must not allow the client to set customerId, createdAt, internal status flags, audit fields, or other server-controlled data.
- Tests cover: a null request field, a maximum length boundary, nested object validation with @Valid, mapping that doesn't leak internal fields, an invalid DTO that never reaches the service, and a stable error code with correlation ID.

KEY TAKEAWAY Well-designed DTOs and effective validation ensure data quality and a trustworthy API contract boundary.

DTO & API CONTRACT: DEFINITION OF DONE

Use this checklist to confirm a DTO and API contract are ready to ship.

- 1 Request and response models are separate where responsibilities differ.
- 2 Only consumer-required fields are exposed.
- 3 Server-controlled fields cannot be over-posted.
- 4 Validation rules are explicit and tested.
- 5 Structural and business validation are separated.
- 6 Mapping is intentional and asymmetric.
- 7 Errors use stable codes and safe messages.
- 8 Nullability and optionality are documented.
- 9 Versioning and compatibility impact are reviewed.
- 10 Examples and schema documentation match implementation.
- 11 Correlation IDs support troubleshooting.
- 12 Sensitive/internal fields never leave the boundary.

KEY TAKEAWAY Run every DTO and API contract through this checklist before it ships.

KNOWLEDGE CHECK (1 OF 2)

Test your understanding of DTOs and Jakarta Validation.

1. What's the primary purpose of a DTO?

- A. To represent database tables
- B. To carry data between client and server**
- C. To handle business logic
- D. To manage transactions

3. Which annotation ensures a String isn't blank?

- A. @NotEmpty
- B. @NotNull
- C. @NotBlank**
- D. @Required

2. Which annotation ensures a field is not null?

- A. @NotNull**
- B. @NotEmpty
- C. @NotBlank
- D. @Size

4. Which annotation validates email format?

- A. @Email**
- B. @Pattern
- C. @ValidEmail
- D. @Format

KEY TAKEAWAY DTOs carry data across boundaries; @NotNull, @NotBlank, and @Email are core Jakarta Validation annotations.

KNOWLEDGE CHECK (2 OF 2)

Four more questions, plus the full answer key.

5. How is validation triggered in a standalone app?

- A. Add @Valid to the controller method
- B. Obtain a Validator from ValidatorFactory and call validate()**
- C. Enable spring.validation.enabled
- D. Annotate the class with @Validated

7. Which annotation validates a collection's size?

- A. @Count
- B. @Size**
- C. @Range
- D. @Length

ANSWER KEY

- 1-B 2-A 3-C 4-A 5-B 6-B 7-B 8-B

6. Does @Email make a field required?

- A. Yes, it implies @NotNull
- B. No — combine with @NotBlank or a presence constraint**
- C. Only when combined with @Size
- D. Yes, but only for String fields

8. What's the typical status code for validation errors?

- A. 200 OK
- B. 400 Bad Request**
- C. 401 Unauthorized
- D. 500 Internal Server Error

KEY TAKEAWAY Use the status your organization's API standard defines — 400 is common for validation, some APIs use 422; consistency matters more than one universal answer.

MODULE 14 COMPLETE

DTOs, Validation and API Contracts

You can now design DTOs, apply Jakarta Validation, and define clear, versioned API contracts with confidence.